

MIGUEL AZEVEDO ALVES

Data & AI Engineer

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PROFILE

MSc Computer Science and Engineering graduate (Instituto Superior Técnico, University of Lisbon) specialised in Artificial Intelligence and Data Science, with an Erasmus exchange at Lund University (Sweden). Thesis compared a rule-based platform against 13 locally-run agentic AI architectures for automating academic-mobility document workflows, shipping a production system rated "Excellent" on usability (SUS 89.88/100, 20 users) alongside a rigorous empirical evaluation of LLM-based document filling and reading. Looking to apply this experience to production Data/AI Engineering roles in Europe.

EXPERIENCE

- Researcher - INESC-ID

Jun 2025 – Set 2026

 - Designed and benchmarked 13 locally-run agentic AI architectures (3 routing, 6 document-filling, 4 reading agents) for automated bureaucratic document processing, using LLMs (Qwen2.5, Llama 3.2) served via Ollama to keep sensitive institutional data on-premise.
 - Achieved 81.3% field accuracy with the best-performing filling agent; found that using semantically descriptive field identifiers over raw PDF field names improved accuracy by up to 30.6 percentage points, and tuned the recommended pipeline to ~100s average end-to-end execution time per process.
 - Built and deployed a rule-based web platform (HTML/JS, PHP, MySQL) integrated with the university's Fénix API and SSO, replacing manual academic-mobility document workflows - achieved a System Usability Scale score of 89.88/100 ("Excellent") in a study with 20 real users; now live in production use.
- Math Tutor - Clubescola

Mar 2024 – Jun 2025

 - Delivered one-on-one mathematics tutoring, tailoring lesson plans to individual learning styles and measurably improving exam performance.
- Embedded Software & Ground Station - MaxPlorer - Aerospace Rescue Team (CANSAT 2021)

2020 – 2021

 - Developed embedded C software and operated the mission ground station, handling real-time data reception, monitoring and analysis.

EDUCATION

- MSc, Computer Science and Engineering - Instituto Superior Técnico, Lisbon - AI & Data Science specialization

2024 – 2026
- Erasmus Exchange, Computer Science and Engineering - Lund University, Sweden

Autumn 2025
- BSc, Computer Science and Engineering - Instituto Superior Técnico, Lisbon

2021 – 2024

TECHNICAL SKILLS

AI / ML: Agentic AI architectures, LLMs (Qwen2.5, Llama 3.2), Ollama, prompt engineering, RAG, PyTorch, scikit-learn
Data: Apache Spark, Pandas, NumPy, data cleaning, transformation and feature engineering, SQL / MySQL
Engineering: Python, Java, JavaScript, PHP, C/C++, REST APIs, Git/GitHub/GitLab, Agile/Scrum/Jira
Cloud / Infra: AWS, Docker

LANGUAGES

Portuguese (Native) · English (Fluent) · French (A1) · Swedish (A1)

VOLUNTEERING

- Scout – Corpo Nacional de Escutas (2012–2025): built leadership, teamwork and community-organisation skills over a decade of active involvement.
- Lead Volunteer – Web Summit Lisbon (2023): coordinated volunteer teams to support event operations.